

MATTHEW HEMA

Mechanical Engineering Student · 3D Modelling & Hardware Prototyping
Computer Vision and Artificial Intelligence · Health & Sustainability Tech

CONTACT

80135814
1010274@mymail.sutd.edu.sg
Nationality: Thai · Age: 20

EDUCATION

B.Eng, Engineering Product Development
SUTD (ASEAN Scholarship)
Major in Mechanical Engineering
Intended Minor: Analytics and AI
Sep 2025 – Present

Global Immersion Programmes
DGIT, Zhejiang University, China (Sep 2026 – Present)
DGIS, Chongqing University, China (Aug 2026)

High School Diploma
Mahidol Wittayanusorn School
Full Scholarship, GPAX 3.94
May 2022 – Mar 2025
IELTS 7.5 (all bands 7.0+) · SAT Math 800

AWARDS & HONOURS

Gold Medal, World Invention Competition and Exhibition (WICE) 2024, Malaysia
Gold Award, Innovation and Design Exchange (IDEX) 2024, Singapore
Winner, SUTD Competitive Programming Contest (Jul 2026)
1st Place, S-Hackathon (IFMSA)
1st Place, ThEP Hackathon (Thailand Center of Excellence in Physics)
2nd Place in Track, Brain Hackathon (PMU)
Best Project Proposal Mention, DGIS, Chongqing University (2026)

TECHNICAL SKILLS

CAD & 3D Modelling (Onshape)
Electronics & Hardware Prototyping
Computer Vision (OpenCV, MediaPipe)
Deep Learning (CNN, LSTM)
Market Research & Project Management

LANGUAGES

Thai (Native) · English (Fluent)
Chinese (Learning)

INTERESTS

Basketball, guitar, videography

SUMMARY

Mechanical engineering student at SUTD who enjoys turning ideas into working hardware. Interested in CAD, computer vision and AI, especially for health and sustainability projects, and always keen to learn new tools and work with people from different backgrounds.

WORK & RESEARCH EXPERIENCE

R&D Intern, BLACKLYTE Sep 2026 – Present
Hangzhou, Zhejiang, China
- Contribute to R&D design of a new product ahead of its market launch
- Conduct market research to shape the product's features and positioning for launch

Research Assistant, Neuroscience Center for Research and Innovation 2024 – 2025
King Mongkut's University of Technology Thonburi (KMUTT)
- MRI-based classification of dementia stages using deep convolutional networks
- Real-time sign language translation system using LSTM neural networks

LEADERSHIP & ACTIVITIES

Player & Content Creator, SUTD Basketball Team and Club 2025 – Present
- Compete for SUTD and create social media content and promotional publications for games and club events

Content Creator, Association of Thai Students in Singapore (ATSIS) Aug 2026 – Present
- Produce social media content and promotional publications for the Thai student community in Singapore

Treasurer, Beauty of Thai SUTD Apr 2026 – Present
- Manage the club's budget and help organise Thai cultural events at SUTD, including Thai language workshops

PROJECTS

Robotic Repositioning System for Pressure-Injury Prevention 2026
- Proposed a robotic system that repositions patients to prevent pressure injuries; Best Project Proposal mention at SUTD DGIS, Chongqing University

Basketball Movement Analysis with Computer Vision 2026 – Present
- Developing a computer vision system that analyses basketball players' movement and tracks their progress

DayCharge: Solar Cell Bicycle Light 2026
- Rebuilt a salvaged bike light into a self-charging solar system (TP4056 module) with a custom Onshape enclosure for handlebar mounting

LazerGraft (Air Draw): Mid-Air Drawing with Hand-Landmark Tracking 2026
- Real-time webcam tool for mid-air drawing, using MediaPipe hand tracking and OpenCV stroke rendering

Solar Food Preservation Device 2026
- Co-designed a solar-powered food dehydrator, handling the mechanical/CAD design and solar electronics

EDUBin: AI-Powered Recycling Educational Tool 2025
- Interactive recycling bin that recognises and sorts items with a deep learning classifier and gives students real-time feedback

Agrisense: Chemical Detection Apparatus for Agriculture and Environment 2024
- Colorimetric device measuring phosphate and nitrate in soil and water; Gold Award, IDEX 2024 and Gold Medal, WICE 2024

VERDIAN-02: Vegetation Remote Detection & Agricultural Satellite 2024
- Designed a CubeSat concept to monitor crops and environmental conditions remotely